

Effect of cooking and germination on phenolic composition and biological properties of dark beans (*Phaseolus vulgaris* L.).



Legumes are the basés diet in several countries. They hold a high nutritional value, but other properties related to human health are nowadays being studied. The aim of this work was to study the influence of processes (boiling or germination) on the phenolic composition of dark beans (*Phaseolus vulgaris* L. c.v. Tolosana) and their effect on their antioxidant, neuroprotective and anticancer ability. Phenolic composition of raw and processed dark beans was analysed by HPLC-PAD and HPLC-ESI/MS. The antioxidant activity was evaluated by ORAC. Astrocytes cultures (U-373) have been used to test their neuroprotective effect. Anticancer activities were evaluated on three different cell lines (renal adenocarcinoma (TK-10), breast adenocarcinoma (MCF-7) and melanoma (UACC-62)) by sulphorhodamine B method. Qualitative and quantitative differences in phenolic composition have been observed between raw and processed dark beans that influence the antioxidant activity, mainly for germinated samples which show a decrease of antioxidant capacity. Although every assayed extracts decreased reactive oxygen species release and exhibited cytotoxicity activities on cancer cell lines, raw beans proved to be the most active in neuroprotective and antitumoral effects; this sample is especially rich in phenolic compounds, mainly anthocyanins. This study further demonstrated that phenolic composition of dark beans is related with cooking process and so with their neuroprotective and anticancer activity; cooking of dark beans improves their digestion and absorption at intestinal level, while maintaining its protective ability on oxidative process at cellular level. Copyright © 2012 Elsevier Ltd. All rights reserved.